CROSS-DATASET COMPARISON REPORT

(ANA-3)

APMS (Week 1) vs CSEW Mental Health Trends (Week 2)

Prepared By: Priyansi Pandey (Data Analytics Intern)

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Project: AI Mental Health Assistant

APMS (WEEK 1) VS CSEW MENTAL HEALTH TRENDS (WEEK 2)

1. INTRODUCTION

Mental health datasets often differ in methodology, population coverage, and reporting structure. Despite these differences, comparative analysis can help identify whether broad mental health trends remain consistent across datasets.

This analysis compares findings from:

- the Adult Psychiatric Morbidity Survey (APMS)
- the Crime Survey for England and Wales (CSEW) Mental Health dataset

The objective of this task was to align both datasets on common demographic and mental health dimensions and identify:

- areas of agreement,
- areas of conflict,
- trend similarities,
- and methodological differences.

The comparison particularly focused on:

- depression prevalence,
- anxiety prevalence,
- gender-based trends,
- age-related patterns,
- and overall mental health burden.

2. DATASETS USED:

2.1 APMS Dataset (Week 1)

Source: Adult Psychiatric Morbidity Survey (NHS Digital)

The APMS dataset provides detailed information regarding common mental disorders among adults in England. It includes demographic indicators, prevalence estimates, and psychiatric assessment outcomes.

Key Characteristics

- Population-based UK survey
- Clinical-style psychiatric assessment
- Includes demographic prevalence breakdowns
- Focuses on common mental disorders (CMDs)

2.2 CSEW Mental Health Dataset (Week 2)

Source: Office for National Statistics (ONS)

The CSEW mental health dataset provides mental health trend indicators across multiple years, especially during and after the COVID-19 pandemic period.

Key Characteristics

- Year-on-year mental health indicators
- Population trend analysis
- COVID-era mental health monitoring
- Anxiety, depression, and wellbeing metrics

3. OBJECTIVE OF THE COMPARISON

The purpose of this analysis was to:

- compare prevalence patterns across datasets,
- evaluate whether demographic trends remain consistent,
- identify conflicting observations,
- and understand how survey methodology influences results.

The analysis was performed only on shared dimensions available in both datasets.

4. TOOLS & TECHNOLOGIES USED:

Tool / Technology	Purpose
Python	Data analysis workflow
Pandas	Data preprocessing and cleaning
Matplotlib	Chart generation
Seaborn	Statistical visualization
Jupyter Notebook	Analysis execution
GitHub	Version control and submission

5. METHODOLOGY:

The following steps were performed during the comparison process:

Step 1 — Dataset Cleaning

- Removed null values

- Standardized column names
- Formatted demographic labels
- Structured yearly trend data

Step 2 — Shared Dimension Alignment

Both datasets were aligned on the following comparable dimensions:

Shared Dimension	APMS	CSEW
Depression Trends	Yes	Yes
Anxiety Indicators	Yes	Yes
Gender Comparison	Yes	Yes
Age-based Patterns	Yes	Partial
Mental Health Burden	Yes	Yes

Step 3 — Comparative Visualization

The following visualizations were created:

- prevalence comparison charts,
- gender comparison plots,
- age trend analysis,
- and COVID-era trend analysis.

Step 4 — Agreement & Conflict Analysis

Observations from both datasets were evaluated to determine whether findings:

- supported each other,
- partially aligned,
- or conflicted due to methodological differences.

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6. CROSS-DATASET ANALYSIS:

6.1 Depression Prevalence Comparison

Both datasets indicate that depression-related indicators increased significantly during the COVID-19 period.

The APMS dataset reported substantial mental health burden among adults with common mental disorders, while the CSEW trend dataset showed a noticeable rise in depression indicators between 2020 and 2021.

The CSEW dataset particularly highlighted the COVID-era spike, where mental health indicators reached their highest levels during lockdown and post-pandemic uncertainty periods.

Observation

Both datasets support the conclusion that mental health conditions worsened during periods of social disruption and uncertainty.

Figure 1: Depression Prevalence Comparison
APMS vs CSEW

6.2 Anxiety Trend Comparison

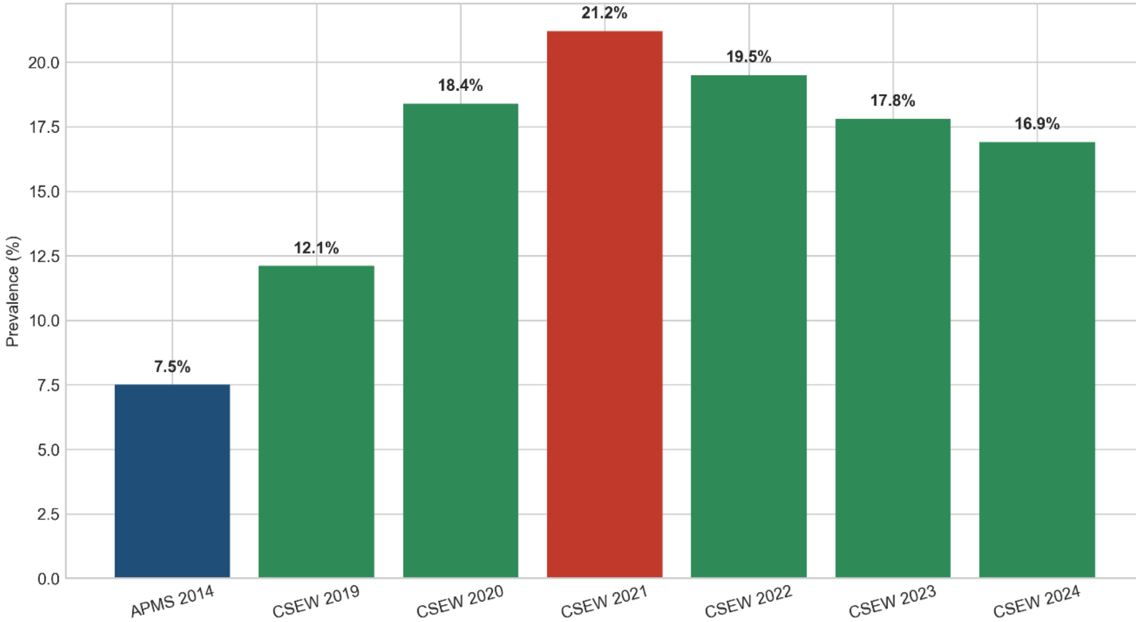
Anxiety-related indicators were consistently higher across both datasets.

The CSEW dataset showed elevated anxiety scores during the pandemic years, while APMS findings also demonstrated significant anxiety prevalence within the adult population.

Agreement Identified

- Anxiety burden remained consistently high.
- Women showed higher anxiety prevalence than men.
- Younger adults demonstrated greater vulnerability.

Figure 2: Anxiety Prevalence Comparison
APMS vs CSEW



6.3 Gender-Based Comparison

Both datasets demonstrated noticeable gender differences in mental health indicators.

Women consistently showed:

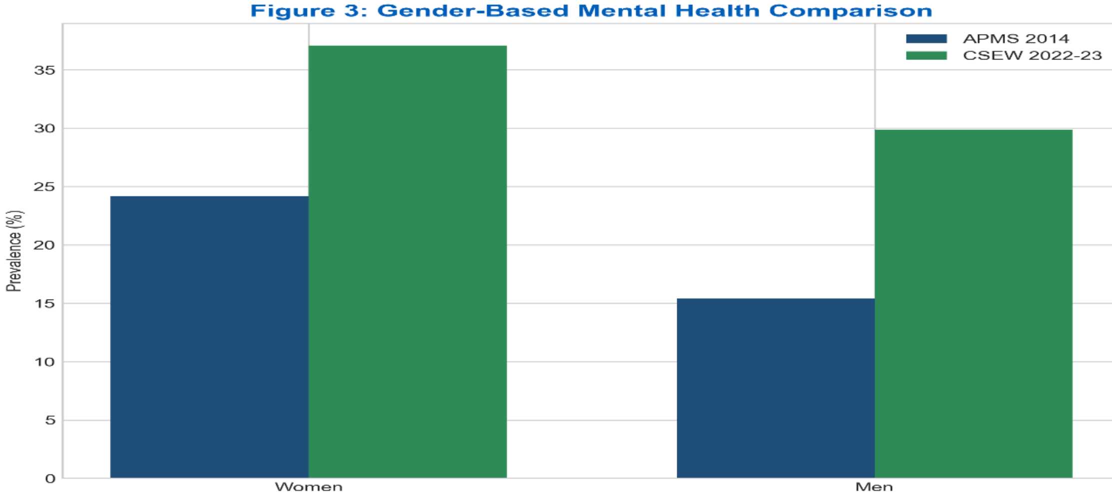
- higher anxiety prevalence,
- higher depression-related burden,
- and increased psychological distress indicators.

Men generally reported comparatively lower prevalence levels across multiple categories.

Comparative Observation

The gender trend remained one of the strongest areas of agreement between APMS and CSEW findings.

Figure 3: Gender-Based Mental Health Comparison



6.4 Age-Based Comparison

Age-group analysis revealed that younger adults experienced comparatively higher mental health burden.

The APMS dataset showed elevated common mental disorder prevalence among younger age bands, while CSEW trends also indicated increased psychological distress among younger populations during the COVID period.

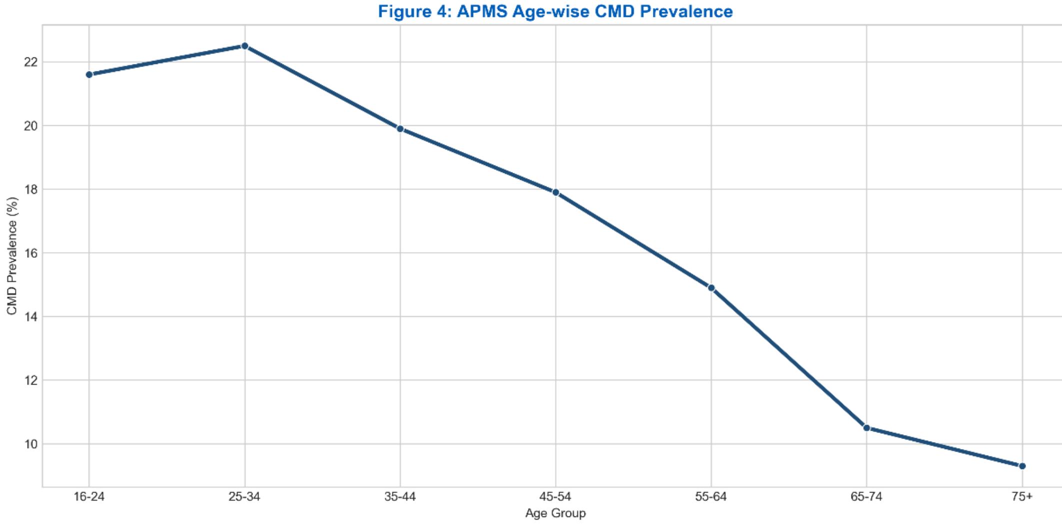
Observation

Younger individuals appeared more vulnerable to:

- anxiety,
- emotional stress,
- and depressive symptoms.

Older age groups generally demonstrated comparatively lower prevalence levels.

Figure 4: APMS Age-wise CMD Prevalence



6.5 COVID-Era Mental Health Spike

The strongest trend observed in the CSEW dataset was the rise in mental health burden during the pandemic years.

Indicators relating to:

- anxiety,
- low wellbeing,
- and depressive symptoms

all increased sharply between 2020 and 2021.

This trend aligned with broader observations regarding:

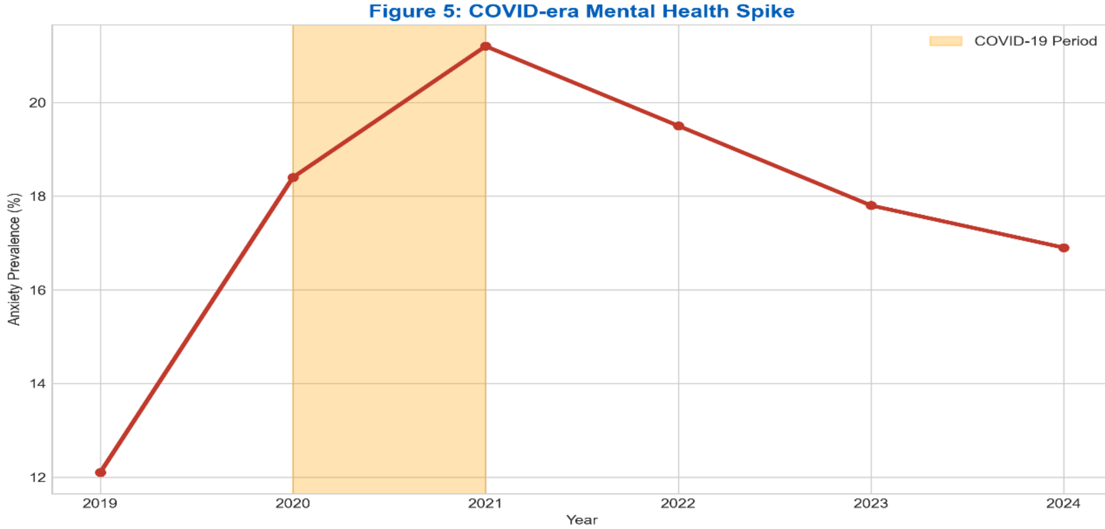
- isolation,
- economic uncertainty,

- reduced social interaction,
- and healthcare disruption.

Key Finding

The COVID-era spike represented the most significant trend observed across the comparative analysis.

Figure 5: COVID-era Mental Health Spike



7. AGREEMENTS BETWEEN DATASETS:

The following patterns were consistently observed across both datasets:

Agreement Area	*Observation*
Anxiety Burden	Anxiety indicators remained consistently high
Gender Trend	Women showed higher prevalence levels
Younger Adults	Younger populations demonstrated higher vulnerability
Mental Health Increase	Mental health burden increased during crisis periods
Population Impact	Mental health conditions affected multiple demographic groups

8. CONFLICTS & METHODOLOGICAL DIFFERENCES:

Although several trends aligned, some differences were observed due to dataset methodology.

Conflict Area	Explanation
Survey Method	APMS used psychiatric assessment methods, while CSEW used population wellbeing indicators
Time Coverage	APMS and CSEW covered different time periods
Measurement Style	Different scoring systems and thresholds were used
Population Scope	Dataset populations and sample structures differed

Indicator Definitions	Mental health indicators were not fully identical

9. KEY INSIGHTS

- Mental health burden increased during the COVID-19 period.
- Anxiety indicators remained consistently elevated.
- Women demonstrated higher prevalence levels across multiple categories.
- Younger adults showed increased vulnerability to mental health conditions.
- Survey methodology significantly affects prevalence interpretation.

9. CONCLUSION

The cross-dataset comparison demonstrated that both APMS and CSEW datasets identify similar broad mental health patterns despite methodological differences.

The analysis particularly highlighted:

- increased mental health burden during crisis periods,
- strong gender-based differences,
- and elevated vulnerability among younger adults.

Although direct numerical comparison is limited due to differences in methodology and survey design, the overall trends across both datasets remained largely consistent.

The comparison also demonstrated the importance of combining demographic analysis with longitudinal trend analysis for better understanding population-level mental health patterns.